1.4 Indices

Indices are also known as powers or exponents. An index tells us how many times a quantity is multiplied by itself. For example,

$$3^4 = 3 \times 3 \times 3 \times 3 = 81$$

In the expression

$$3^4$$

the number 3 is called the base and the number 4 is called the index. Indices provide a compact way to write repeated multiplication. For example,

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

can be written more simply as

$$2^8$$

Definition

If no index is shown, the index is 1. For example

$$5 = 5^1$$

and

$$x = x^1$$

1.4.1 Multiplying Indices

Consider

$$4^3 \times 4^5$$

Writing each power in full gives

Counting the factors of 4 shows that there are eight in total:

Notice that

This leads to the following rule.

Definition

When multiplying powers with the same base, add the indices.

$$a^m \times a^n = a^{m+n}$$

Worked Example

Simplify

$$2^4 \times 2^7$$

Add the indices:

1.4.2 Dividing Indices

Consider

$$\frac{6^5}{6^2}$$

Writing each power in full gives

Two factors of 6 cancel, leaving

Notice that

This leads to the following rule.

Definition

When dividing powers with the same base, subtract the indices.

$$\frac{a^m}{a^n} = a^{m-n}$$

Worked Example

Simplify

$$\frac{7^8}{7^3}$$

Subtract the indices:

1.4.3 Powers of Powers

Consider the expression

$$(a^3)^2$$

Writing the inner power in full gives

This means

Collecting all the factors gives

Since

we obtain the following rule.

Definition

When a power is raised to another power, multiply the indices.

$$(a^m)^n = a^{mn}$$

Worked Example

Simplify

$$(5^2)^3$$

Multiply the indices:

1.4.4 Zero Indices

Consider

$$\frac{a^2}{a^2}$$

Using the division rule,

However,

Therefore,

This gives the following rule.

Definition

Any non-zero number raised to the power zero is equal to one.

$$a^0 = 1$$

Worked Example

Evaluate 17^0.

$$17^0 = 1$$

1.4.5 Negative Indices

Many students think a negative index produces a negative answer. This is not true. A negative index indicates a reciprocal. Consider

$$\frac{a^2}{a^6}$$

Using the division rule,

However,

Therefore,

This leads to the following rule.

Definition

A negative index indicates a reciprocal.

$$a^{-m} = \frac{1}{a^m}$$

Worked Example

Evaluate 2^{-3}.

Using the rule,

$$2^{-3} =$$

1.4.6 Fractional Indices

Consider

$$a^{\frac{1}{2}} \times a^{\frac{1}{2}}$$

Using the multiplication rule,

We also know that

Therefore,

This leads to the following rule.

Definition

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Worked Example

Evaluate $64^{\frac{1}{3}}$.

$$64^{\frac{1}{3}} =$$

More generally,

$$(a^{1/n})^m = a^{m/n}$$

Hence,

Definition

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

Worked Example

Evaluate $25^{\frac{3}{2}}$.

$$25^{\frac{3}{2}} =$$

1.4.7 Common Mistakes

Common errors involving indices include:

- Thinking $a^m + a^n = a^{m+n}$.
- Confusing $(a^m)^n$ with $a^m \times a^n$.
- Assuming a negative index gives a negative answer.
- Forgetting that $a^0 = 1$.
- Forgetting that fractional indices represent roots.
- Ignoring brackets, for example confusing $(-2)^4$ with -2^4.

1.4.8 Engineering Application

Engineering Application

Indices occur frequently throughout engineering and science.

The area of a square plate with side length 3 m is

$$3^2 = 9 \text{ m}^2$$

The volume of a cube with side length 2 m is

$$2^3 = 8 \text{ m}^3$$

Indices also appear in scientific notation. For example,

$$3 \times 10^6$$

represents three million.

Understanding indices is therefore essential when working with formulae, measurements and engineering calculations.

Summary

Definition

For any suitable values of a, m and n,

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^0 = 1$$

$$a^{1/n} = \sqrt[n]{a}$$

$$a^{m/n} = \sqrt[n]{a^m}$$

Indices appear throughout mathematics, engineering and science. These rules should become second nature.

1.4.9 Exercises

Exercise

1. Evaluate:
   (a) 7^3
   (b) 3^7
   (c) 5^{-2}
   (d) $2^{5/2}$
   (e) $(-2)^6$
   (f) -2^6

2. Express each as a single power:
   (a) $2^3 \times 2^5 \times 2^7$
   (b) $3^3 \times 3^{-4}$
   (c) $4^6 \times \dfrac{4^{-7}}{4^{-5}}$
   (d) $\dfrac{5^5 \times 5^{-3}}{5^4 \times 5^{-2} \times 5^{-7}}$
   (e) $(2^{-3})^4$
   (f) $3^{2/3} \times 3^{-2}$

3. Simplify:
   (a) $y^5 \times y^8$
   (b) $\dfrac{x^{10}}{x^7}$
   (c) $x^3 \div x$
   (d) $(a^2)^3$
   (e) $3x^2 \times 4x^7 \times 2x^{-3}$
   (f) $6x \times 7y^2 \times x^5$

4. Change to the specified base:
   (a) 25^3 to base 5
   (b) 8^6 to base 2
   (c) 9^4 to base 3
   (d) 81^5 to base 3

Exercise

5. Determine the value of y:

$$16^{1/4} \times 2^y = 8^{3/4}$$

6. Evaluate:

   (a) 10^0

   (b) $81^{1/2}$

   (c) $64^{1/3}$

   (d) $25^{3/2}$

   (e) $\dfrac{3^5 \times 3^2}{3^4}$

7. A patient has 4^3 affected cells on day 1. The number of affected cells doubles every day. The disease becomes serious when 2^{10} cells are affected. On which day does the disease become serious?

8. The area of a rectangle is $125^{1/4}\,\text{cm}^2$. The side lengths are $5^{x+1}\,\text{cm}$ and $25^{1/2}\,\text{cm}$. Determine x.

Solutions to Exercise 1.4.9

1. Evaluate:

   (a) 7^3

$$7^3 = 7 \times 7 \times 7 = 343$$

   (b) 3^7

$$3^7 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 2187$$

   (c) 5^{-2}

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

   (d) $2^{5/2}$

$$2^{5/2} = (\sqrt{2})^5 = 4\sqrt{2}$$

   or

$$2^{5/2} = \sqrt{2^5} = \sqrt{32} = 4\sqrt{2}$$

(e) $(-2)^6$

$$(-2)^6 = 64$$

because an even number of negative signs gives a positive result.

(f) -2^6

The index is applied first.

$$-2^6 = -(2^6) = -64$$

2. Express each as a single power:

(a) $2^3 \times 2^5 \times 2^7$

$$2^{3+5+7} = 2^{15}$$

(b) $3^3 \times 3^{-4}$

$$3^{3-4} = 3^{-1}$$

(c) $4^6 \times \dfrac{4^{-7}}{4^{-5}}$

$$\begin{aligned} 4^6 \times 4^{-7-(-5)} &= 4^6 \times 4^{-2} \\ &= 4^4 \end{aligned}$$

(d)

$$\frac{5^5 \times 5^{-3}}{5^4 \times 5^{-2} \times 5^{-7}}$$

Numerator:

$$5^{5-3} = 5^2$$

Denominator:

$$5^{4-2-7} = 5^{-5}$$

Therefore

$$\frac{5^2}{5^{-5}} = 5^{2-(-5)} = 5^7$$

(e) $(2^{-3})^4$

$$2^{-3\times 4} = 2^{-12}$$

(f) $3^{2/3} \times 3^{-2}$

$$3^{2/3-2} = 3^{2/3-6/3} = 3^{-4/3}$$

3. Simplify:

   (a) $y^5 \times y^8$

$$y^{5+8} = y^{13}$$

   (b) $\dfrac{x^{10}}{x^7}$

$$x^{10-7} = x^3$$

   (c) $x^3 \div x$

$$x^{3-1} = x^2$$

   (d) $(a^2)^3$

$$a^{2\times 3} = a^6$$

   (e) $3x^2 \times 4x^7 \times 2x^{-3}$
   Multiply coefficients:

$$3 \times 4 \times 2 = 24$$

   Combine powers:

$$x^{2+7-3} = x^6$$

   Therefore

$$24x^6$$

   (f) $6x \times 7y^2 \times x^5$

$$42x^{1+5}y^2 = 42x^6y^2$$

4. Change to the specified base:

   (a) 25^3 to base 5.

$$25 = 5^2$$

   Therefore

$$25^3 = (5^2)^3 = 5^6$$

   (b) 8^6 to base 2.

$$8 = 2^3$$

   Therefore

$$8^6 = (2^3)^6 = 2^{18}$$

(c) 9^4 to base 3.

$$9 = 3^2$$

Therefore

$$9^4 = (3^2)^4 = 3^8$$

(d) 81^5 to base 3.

$$81 = 3^4$$

Therefore

$$81^5 = (3^4)^5 = 3^{20}$$

5. Determine the value of y:

$$16^{1/4} \times 2^y = 8^{3/4}$$

Write everything in base 2.

$$\begin{aligned} (2^4)^{1/4} \times 2^y &= (2^3)^{3/4} \\ 2 \times 2^y &= 2^{9/4} \\ 2^{1+y} &= 2^{9/4} \end{aligned}$$

Therefore

$$\begin{aligned} 1 + y &= \frac{9}{4} \\ y &= \frac{9}{4} - 1 = \frac{5}{4} \end{aligned}$$

6. Evaluate:

(a) 10^0

$$10^0 = 1$$

(b) $81^{1/2}$

$$\sqrt{81} = 9$$

(c) $64^{1/3}$

$$\sqrt[3]{64} = 4$$

(d) $25^{3/2}$

$$(\sqrt{25})^3 = 5^3 = 125$$

(e)

$$\frac{3^5 \times 3^2}{3^4} = 3^{5+2-4}$$
$$= 3^3 = 27$$

7. Patient problem Initially:

$$4^3 = 64 = 2^6$$

The number doubles every day. Day 1:

$$2^6$$

Day 2:

$$2^7$$

Day 3:

$$2^8$$

Day 4:

$$2^9$$

Day 5:

$$2^{10}$$

The disease becomes serious on

$$\boxed{\text{Day 5}}$$

8. Rectangle problem Area

$$125^{1/4}$$

Side lengths:

$$5^{x+1} \quad \text{and} \quad 25^{1/2}$$

Simplify the second side:

$$25^{1/2} = 5$$

Therefore

$$5^{x+1} \times 5 = 125^{1/4}$$

Write everything in base 5.

$$5^{x+1} \times 5^1 = (5^3)^{1/4}$$
$$5^{x+2} = 5^{3/4}$$

Hence

$$x + 2 = \frac{3}{4}$$
$$x = \frac{3}{4} - 2$$
$$x = \frac{3}{4} - \frac{8}{4}$$
$$x = -\frac{5}{4}$$
$$x = -\frac{5}{4}$$